

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]

Tap amplitudes (pixels): [129.65338406690356, 129.5414991421668, 115.83177456984762, 110.01818031580054, 116.84605256490268, 107.33592129385204, 95.01578816175763, 116.76472069936193, 116.8289347721702, 114.12712210513327, 116.81181447096864]



Analysis of Extracted Features

To determine the MDS-UPDRS score, we evaluate the data against the three primary criteria: **Speed, Rhythm (Interruptions)**, and **Amplitude (Decrement)**.

1. Speed and Rhythm (Tap Periods)

- **Data:** [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]
- **Trend:** The subject actually **speeds up** over the course of the test. They start at a period of 123 frames and quickly settle into a faster rhythm (averaging ~99 frames for the latter half).
- **Interruptions:** There are no significant spikes in the period data that would suggest a "halt" or "freeze." The variance (e.g., 92 to 118) is within a normal range of human motor variability and does not constitute a rhythm break or hesitation.
- **Conclusion:** No slowing; no interruptions.

2. Amplitude (Tap Amplitudes)

- **Data:** [129.65, 129.54, 115.83, 110.02, 116.85, 107.34, 95.02, 116.76, 116.83, 114.13, 116.81]

- **Trend:** The subject starts with high amplitude (~129 px) for the first two taps. There is a drop to a stable baseline of ~116 px for the remainder of the test.
- **Decrement Analysis:**
 - **Score 1 Criteria (Near End):** The amplitude does *not* decrement near the end. In fact, the last four taps (116 , 116 , 114 , 116) show a very stable recovery and consistency.
 - **Score 2/3 Criteria (Midway/Early):** While there is a drop after the second tap, the amplitude stabilizes rather than continuing to degrade. This "settling in" pattern differs from the pathological "sequence effect" (fatigue) seen in Parkinsonism, where amplitude progressively shrinks. The single dip at tap 7 (95.02 px) is isolated and immediately recovers.
- **Conclusion:** Amplitude is well-maintained with no progressive decrement (fatigue).

Scoring Determination

The subject demonstrates consistent rhythm (actually improving in speed) and maintains a stable amplitude profile after an initial settling period. There are no hesitations, halts, freezes, or pathological decrements indicative of bradykinesia.

Score: 0